Netflix Eda Project

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("netflix_titles.csv")

df.head()
```

```
  show_id     type                   title         director  \
0      s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show          Blood & Water              NaN
2      s3  TV Show              Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans              NaN
4      s5  TV Show           Kota Factory              NaN

                                                   cast        country  \
0                                                   NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...            NaN
3                                                   NaN            NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...          India

           date_added  release_year rating   duration  \
0  September 25, 2021          2020  PG-13     90 min
1  September 24, 2021          2021  TV-MA  2 Seasons
2  September 24, 2021          2021  TV-MA   1 Season
3  September 24, 2021          2021  TV-MA   1 Season
4  September 24, 2021          2021  TV-MA  2 Seasons

                                          listed_in  \
0                                     Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act...
3                             Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV ...

                                           description
0  As her father nears the end of his life, filmm...
1  After crossing paths at a party, a Cape Town t...
2  To protect his family from a powerful drug lor...
3  Feuds, flirtations and toilet talk go down amo...
4  In a city of coaching centers known to train I...
```

```
df.shape
df.columns
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   str
 1   type          8807 non-null   str
 2   title         8807 non-null   str
 3   director      6173 non-null   str
 4   cast          7982 non-null   str
 5   country       7976 non-null   str
 6   date_added    8797 non-null   str
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   str
 9   duration      8804 non-null   str
 10  listed_in     8807 non-null   str
 11  description   8807 non-null   str
dtypes: int64(1), str(11)
memory usage: 825.8 KB
```

```
df.isnull().sum()
```

```
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64
```

--> Found huge number of missing values.

```
df.fillna(({'director' :'Unknown'}), inplace=True)
```

```
     show_id     type                  title         director  \
0         s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1         s2  TV Show          Blood & Water          Unknown
2         s3  TV Show              Ganglands  Julien Leclercq
3         s4  TV Show  Jailbirds New Orleans          Unknown
4         s5  TV Show           Kota Factory          Unknown
...      ...      ...                    ...              ...
8802   s8803    Movie                 Zodiac    David Fincher
8803   s8804  TV Show            Zombie Dumb          Unknown
8804   s8805    Movie             Zombieland  Ruben Fleischer
```

```
8805   s8806    Movie                    Zoom     Peter Hewitt
8806   s8807    Movie                  Zubaan      Mozez Singh

                                                     cast        country
\
0                                                     NaN  United States

1     Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...   South Africa

2     Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...            NaN

3                                                     NaN            NaN

4     Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...          India

...                                                   ...            ...

8802  Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...  United States

8803                                                  NaN            NaN

8804  Jesse Eisenberg, Woody Harrelson, Emma Stone, ...  United States

8805  Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...  United States

8806  Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...          India


              date_added  release_year rating   duration  \
0     September 25, 2021          2020  PG-13     90 min
1     September 24, 2021          2021  TV-MA  2 Seasons
2     September 24, 2021          2021  TV-MA   1 Season
3     September 24, 2021          2021  TV-MA   1 Season
4     September 24, 2021          2021  TV-MA  2 Seasons
...                  ...           ...    ...        ...
8802   November 20, 2019          2007      R    158 min
8803        July 1, 2019          2018  TV-Y7  2 Seasons
8804    November 1, 2019          2009      R     88 min
8805    January 11, 2020          2006     PG     88 min
8806       March 2, 2019          2015  TV-14    111 min

                                                listed_in  \
0                                          Documentaries
1       International TV Shows, TV Dramas, TV Mysteries
2     Crime TV Shows, International TV Shows, TV Act...
3                                Docuseries, Reality TV
4     International TV Shows, Romantic TV Shows, TV ...
...                                                  ...
8802                      Cult Movies, Dramas, Thrillers
8803             Kids' TV, Korean TV Shows, TV Comedies
8804                             Comedies, Horror Movies
```

```
8805                  Children & Family Movies, Comedies
8806     Dramas, International Movies, Music & Musicals

                                                description
0     As her father nears the end of his life, filmm...
1     After crossing paths at a party, a Cape Town t...
2     To protect his family from a powerful drug lor...
3     Feuds, flirtations and toilet talk go down amo...
4     In a city of coaching centers known to train I...
...                                                 ...
8802  A political cartoonist, a crime reporter and a...
8803  While living alone in a spooky town, a young g...
8804  Looking to survive in a world taken over by zo...
8805  Dragged from civilian life, a former superhero...
8806  A scrappy but poor boy worms his way into a ty...

[8807 rows x 12 columns]
```

```
df.fillna(({'cast':'Unknown'}), inplace=True)
df.fillna(df['country'].mode()[0], inplace=True)
df.fillna(df['rating'].mode()[0], inplace=True)
df.fillna(df['duration'].mode()[0], inplace=True)
```

```
     show_id     type                   title         director  \
0         s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1         s2  TV Show          Blood & Water          Unknown
2         s3  TV Show              Ganglands  Julien Leclercq
3         s4  TV Show  Jailbirds New Orleans          Unknown
4         s5  TV Show           Kota Factory          Unknown
...      ...      ...                    ...              ...
8802   s8803    Movie                 Zodiac    David Fincher
8803   s8804  TV Show            Zombie Dumb          Unknown
8804   s8805    Movie             Zombieland  Ruben Fleischer
8805   s8806    Movie                   Zoom     Peter Hewitt
8806   s8807    Movie                 Zubaan      Mozez Singh

                                                   cast        country
\
0                                             Unknown  United States

1     Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...   South Africa

2     Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...  United States

3                                             Unknown  United States

4     Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...          India

...                                                ...            ...

8802  Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...  United States
```

```
8803                                                Unknown  United States

8804  Jesse Eisenberg, Woody Harrelson, Emma Stone, ...  United States

8805  Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...  United States

8806  Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...          India


              date_added  release_year rating   duration  \
0     September 25, 2021          2020  PG-13     90 min
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4     September 24, 2021          2021  TV-MA  2 Seasons
...                  ...           ...    ...        ...
8802   November 20, 2019          2007      R    158 min
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8804    November 1, 2019          2009      R     88 min
8805    January 11, 2020          2006     PG     88 min
8806       March 2, 2019          2015  TV-14    111 min

                                                listed_in  \
0                                          Documentaries
1       International TV Shows, TV Dramas, TV Mysteries
2     Crime TV Shows, International TV Shows, TV Act...
3                                Docuseries, Reality TV
4     International TV Shows, Romantic TV Shows, TV ...
...                                                  ...
8802                      Cult Movies, Dramas, Thrillers
8803             Kids' TV, Korean TV Shows, TV Comedies
8804                            Comedies, Horror Movies
8805                 Children & Family Movies, Comedies
8806     Dramas, International Movies, Music & Musicals

                                               description
0     As her father nears the end of his life, filmm...
1     After crossing paths at a party, a Cape Town t...
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...                                                 ...
8802  A political cartoonist, a crime reporter and a...
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8805  Dragged from civilian life, a former superhero...
8806  A scrappy but poor boy worms his way into a ty...

[8807 rows x 12 columns]
```

```
df.isnull().sum()
```

```
show_id         0
type            0
title           0
director        0
cast            0
country         0
date_added      0
release_year    0
rating          0
duration        0
listed_in       0
description     0
dtype: int64
```

```
df.drop_duplicates()
df.head(1)
```

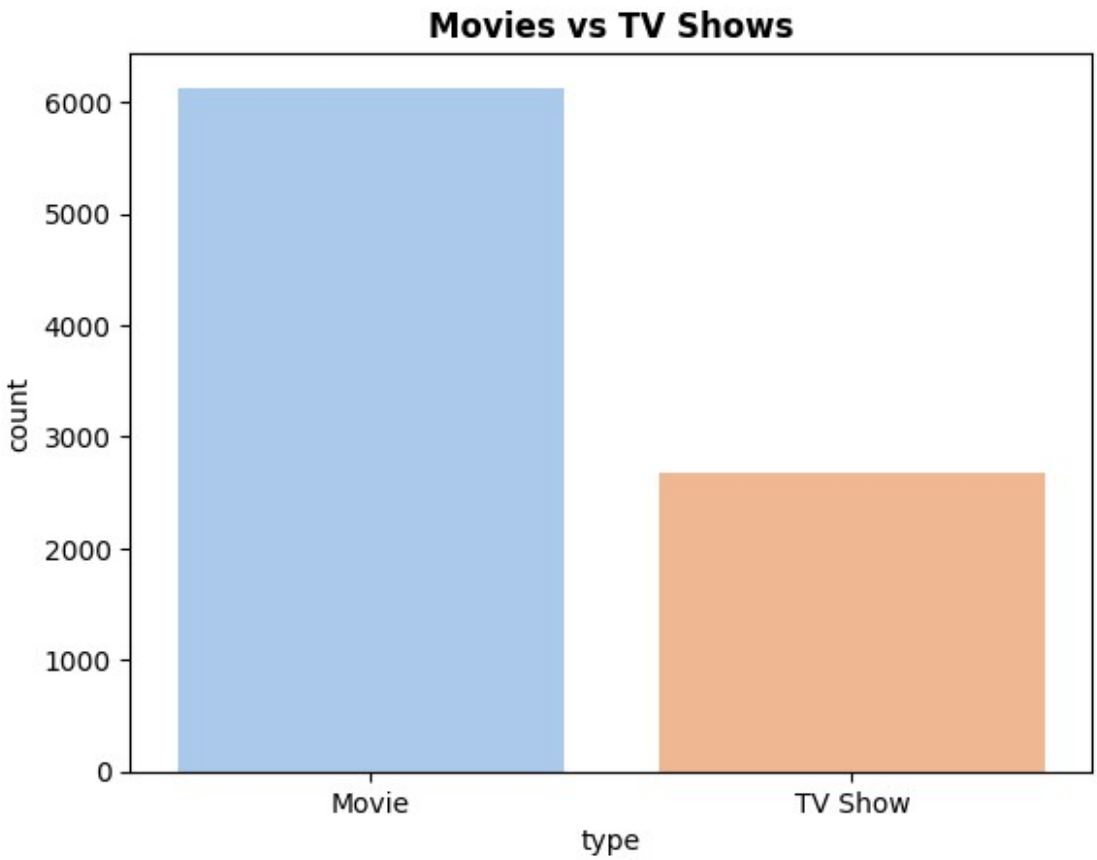
```
  show_id   type                 title         director     cast  \
0      s1  Movie  Dick Johnson Is Dead  Kirsten Johnson  Unknown

         country          date_added  release_year rating duration  \
0  United States  September 25, 2021          2020  PG-13   90 min

       listed_in                                          description
0  Documentaries  As her father nears the end of his life, filmm...
```

```
df.drop("description", axis=1,inplace=True)
df.head(1)
```

```
  show_id   type                 title         director     cast  \
0      s1  Movie  Dick Johnson Is Dead  Kirsten Johnson  Unknown

         country          date_added  release_year rating duration  \
0  United States  September 25, 2021          2020  PG-13   90 min

       listed_in
0  Documentaries
```

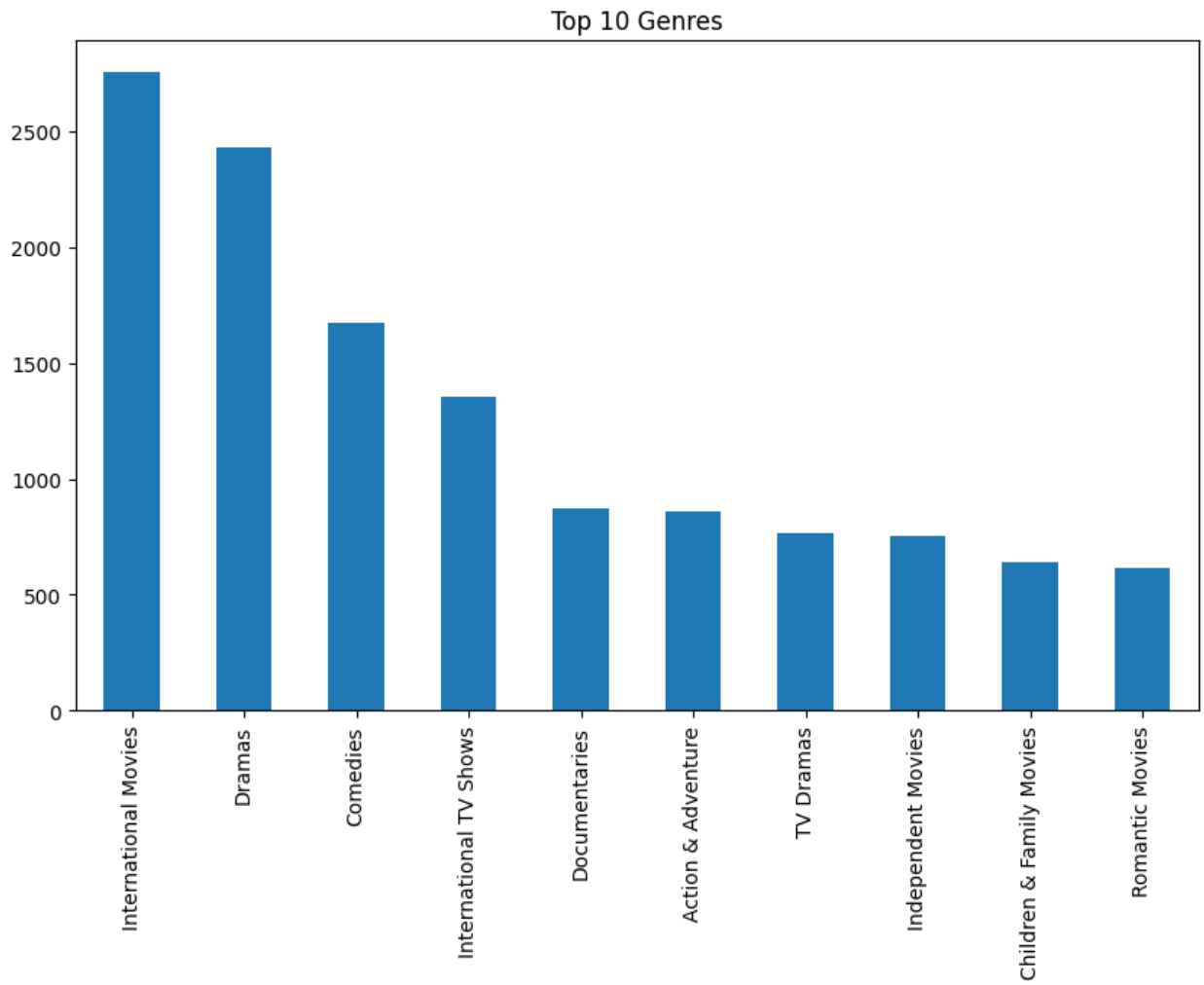
Visualization

```
sns.countplot(data=df, x='type', hue='type' ,palette='pastel')
plt.title("Movies vs TV Shows",fontsize=12, fontweight='bold')
plt.show()
```



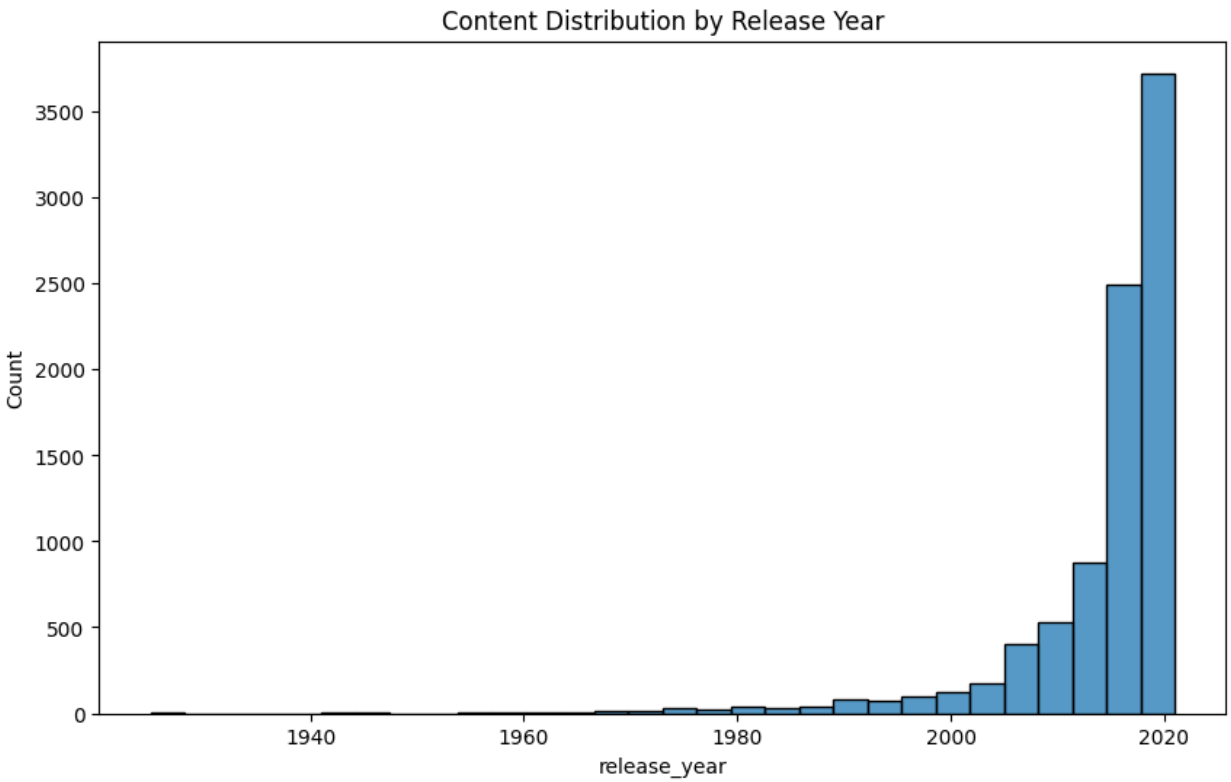
--> There are more than double the number of movies compared to Tv shows.

```
genres = df['listed_in'].str.split(',', expand=True).stack()
genres = genres.str.strip()

plt.figure(figsize=(10,6))
genres.value_counts().head(10).plot(kind='bar')
plt.title("Top 10 Genres")
plt.show()
```

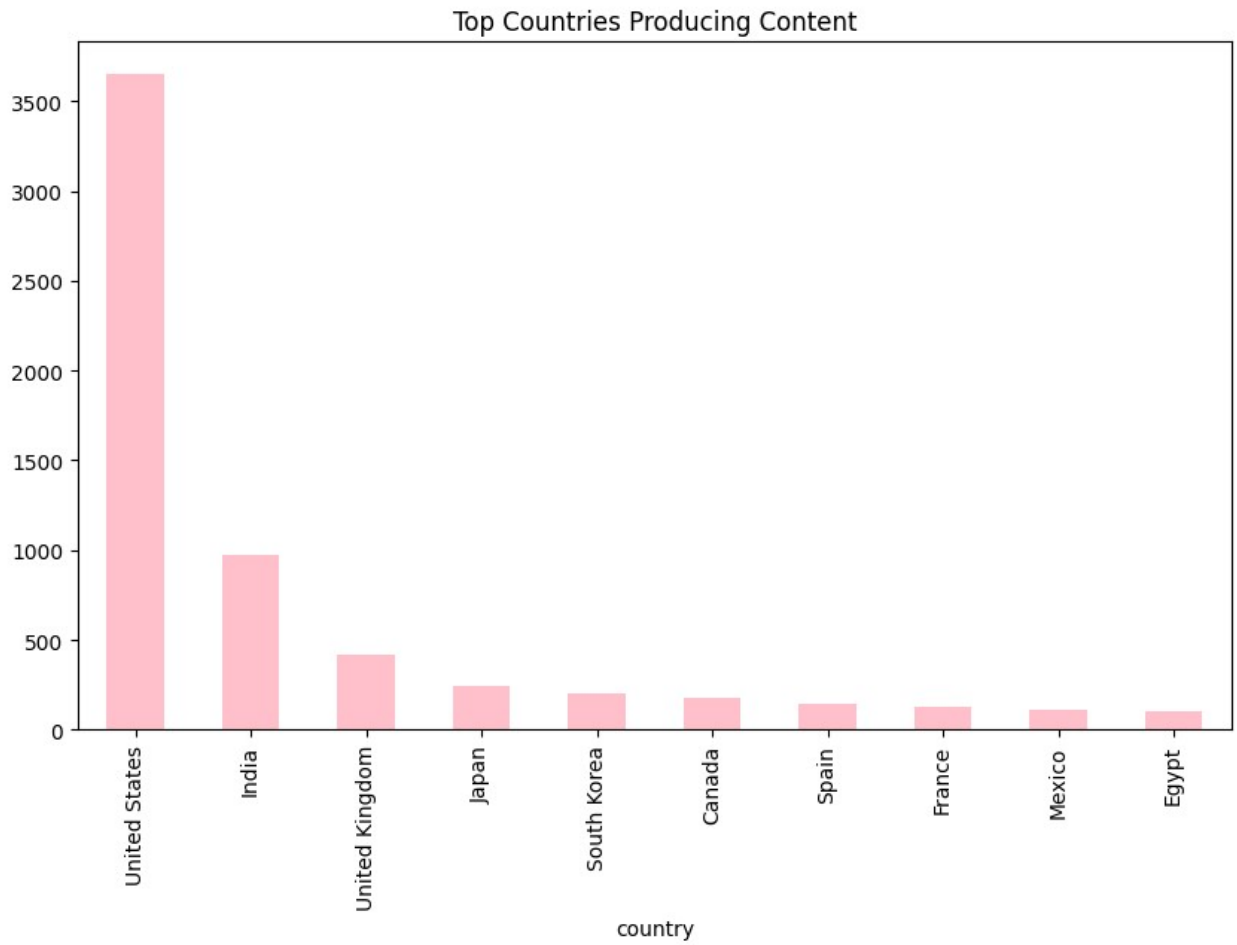


--> International movies is the most common genre. This means the platform has a very large collection of movies from all over the world.

```
plt.figure(figsize=(10,6))
sns.histplot(df['release_year'], bins=30)
plt.title("Content Distribution by Release Year")
plt.show()
```

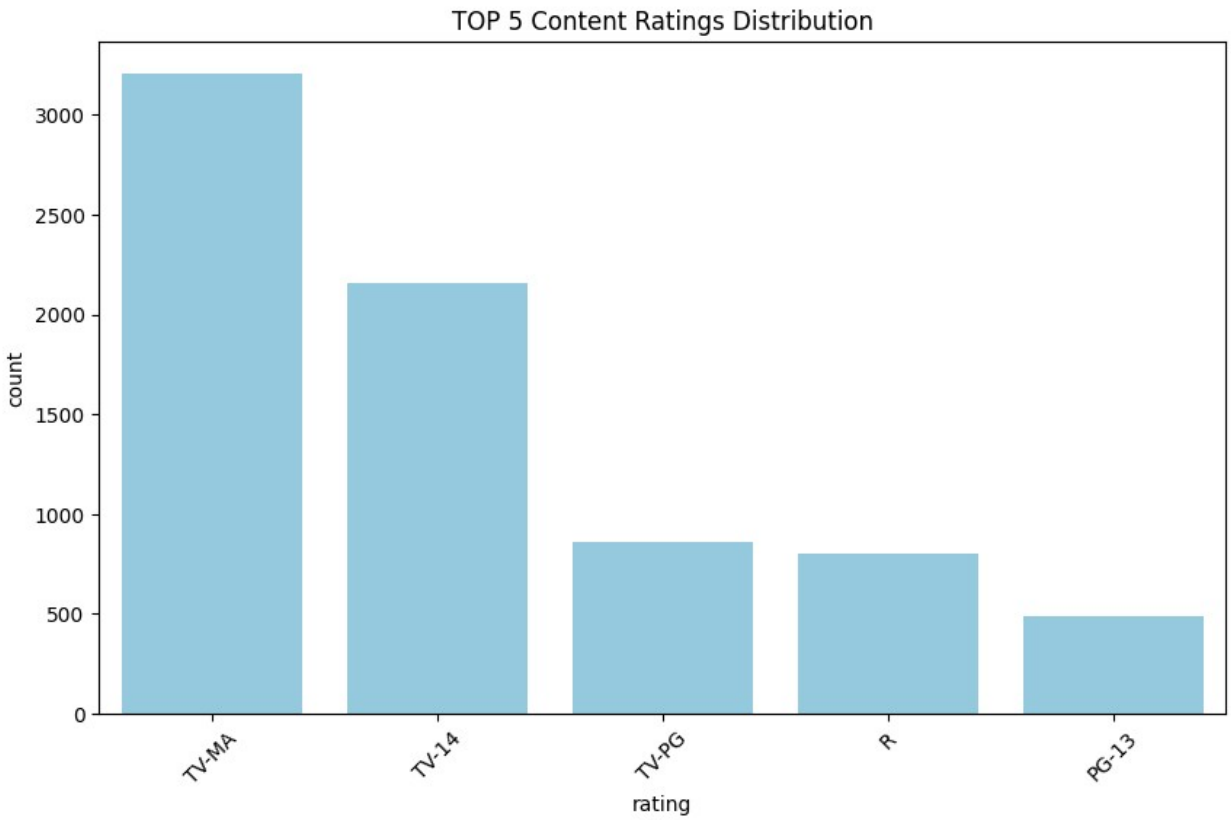



```
top_countries = df['country'].value_counts().head(10)

top_countries.plot(kind='bar', figsize=(10,6), color='pink')
plt.title("Top Countries Producing Content")
plt.show()
```



--> United States has produced the highest content on netflix.

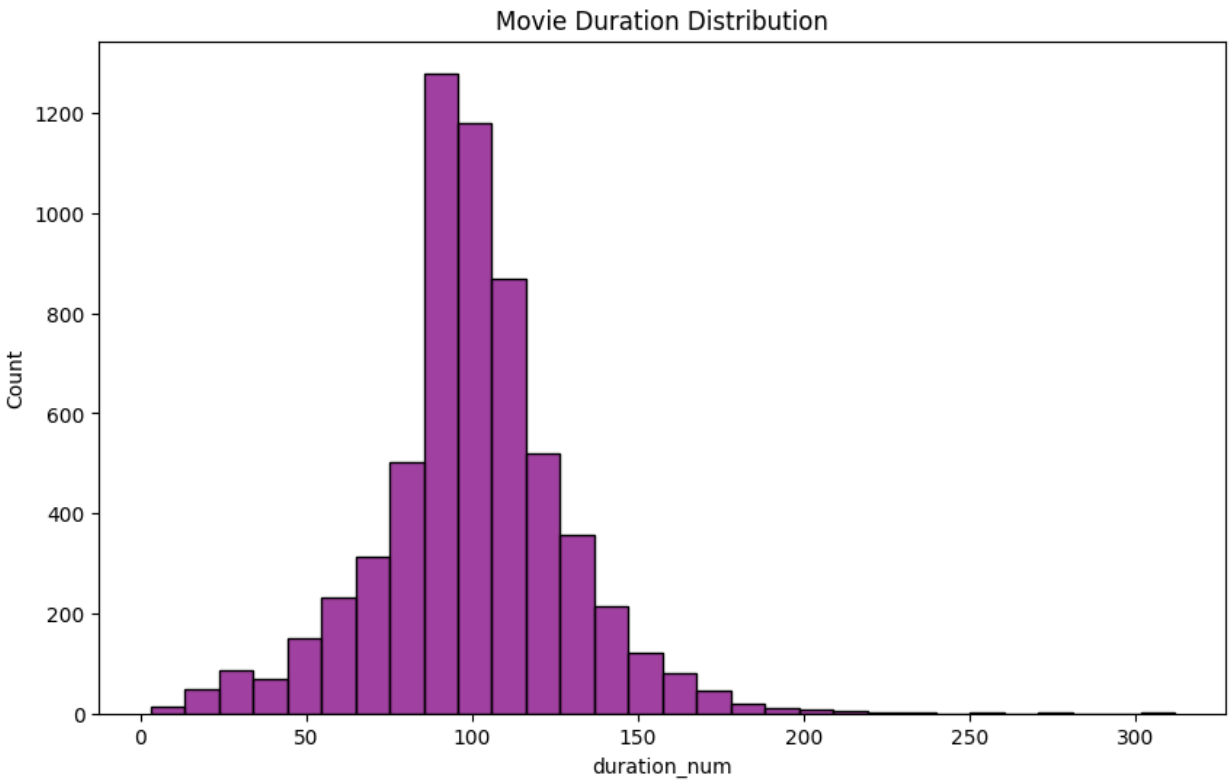
```
plt.figure(figsize=(10,6))
sns.countplot(data=df, x='rating',
order=df['rating'].value_counts().head(5).index, color='skyblue')
plt.xticks(rotation=45)
plt.title("TOP 5 Content Ratings Distribution")
plt.show()
```



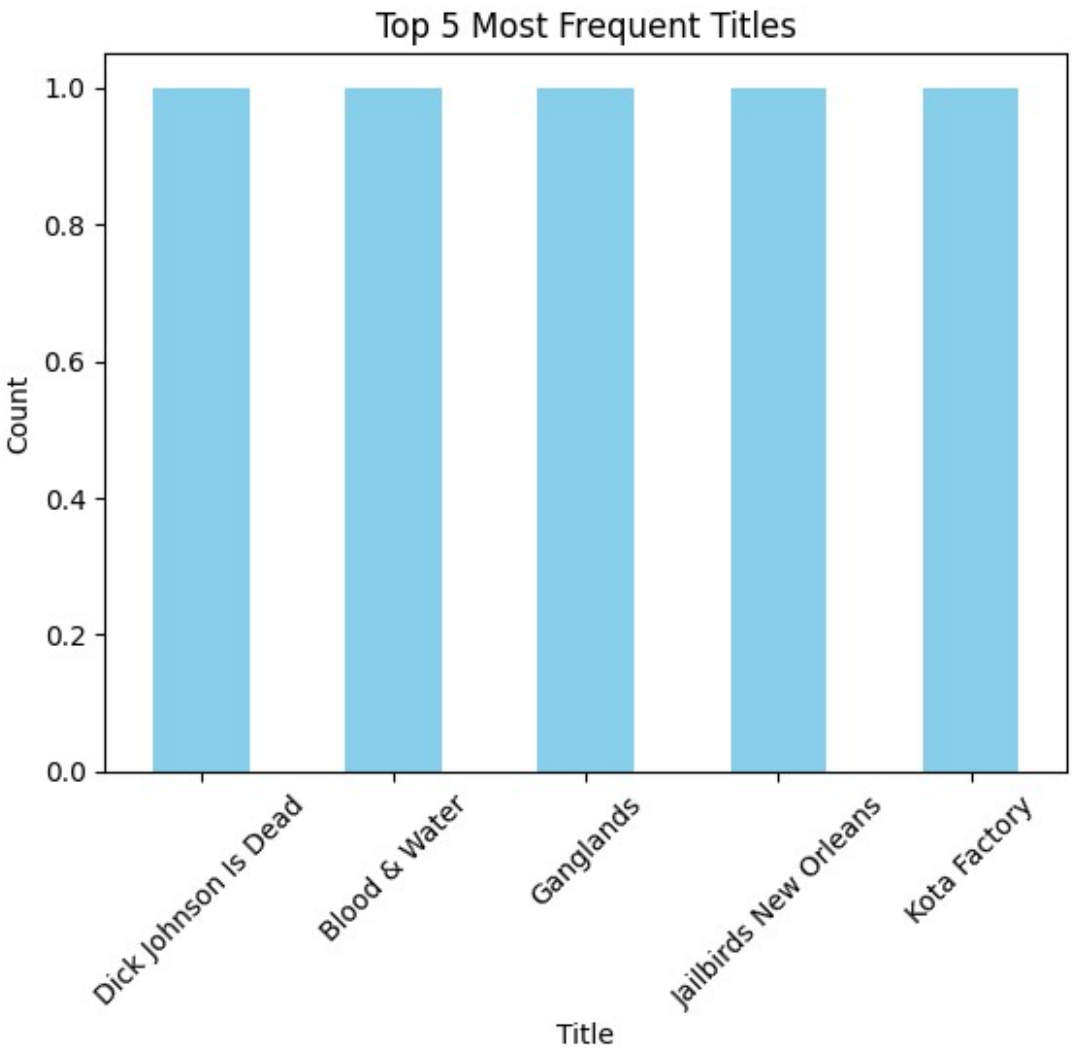
--> MOst of the content on Netflix is rated TV-MA.

```
movies = df[df['type'] == 'Movie']

movies['duration_num'] = movies['duration'].str.extract('(\
d+)').astype(float)

plt.figure(figsize=(10,6))
sns.histplot(movies['duration_num'], bins=30, color='purple')
plt.title("Movie Duration Distribution")
plt.show()
```

```
<>:3: SyntaxWarning: "\d" is an invalid escape sequence. Such
sequences will not work in the future. Did you mean "\\d"? A raw
string is also an option.
<>:3: SyntaxWarning: "\d" is an invalid escape sequence. Such
sequences will not work in the future. Did you mean "\\d"? A raw
string is also an option.
C:\Users\ExpertBook\AppData\Local\Temp\
ipykernel_15712\4150199707.py:3: SyntaxWarning: "\d" is an invalid
escape sequence. Such sequences will not work in the future. Did you
mean "\\d"? A raw string is also an option.
  movies['duration_num'] = movies['duration'].str.extract('(\
d+)').astype(float)
```




```
top_rated_movies = df['title'].value_counts().head(5)

top_rated_movies.plot(kind='bar', color='skyblue')
plt.xticks(rotation=45)
plt.title("Top 5 Most Frequent Titles")
plt.ylabel("Count")
plt.xlabel("Title")
plt.show()
```




```
print("Unique Genres:", df['listed_in'].nunique())
print("Unique Countries:", df['country'].nunique())

Unique Genres: 514
Unique Countries: 748
```

Conclusion

- Netflix has far more Movies than TV Shows, so its catalog is mainly film-focused. The platform is also strongly international, with International Movies, Dramas, and Comedies appearing most often.
- The United States contributes the most titles, followed by India, and most content is rated TV-MA, showing a strong focus on adult audiences. Most titles are from recent years, movies are usually around 100 minutes, and most TV shows have only 1 season.

Recommendations

- Keep investing in movies, since they make up most of the catalog and seem to be Netflix's core strength.
- Expand international content further, especially from strong markets like India, South Korea, Japan, and other growing regions.
- Add more family and kids content, because the catalog is currently more focused on TV-MA and adult audiences.